100 Must-Do DSA Questions

Covering All Major Patterns for Coding Interviews

Notes by Shubham Soni

Pattern 1: Arrays & Hashing

#	Problem Name	Difficulty	LeetCode Link
1	Two Sum	Easy	View Problem
2	Contains Duplicate	Easy	View Problem
3	Valid Anagram	Easy	View Problem
4	Group Anagrams	Medium	View Problem
5	Top K Frequent Elements	Medium	View Problem
6	Product of Array Except Self	Medium	View Problem
7	Longest Consecutive Sequence	Medium	View Problem

Pattern 2: Two Pointers

#	Problem Name	Difficulty	LeetCode Link
8	Valid Palindrome	Easy	View Problem
9	Two Sum II - Input Array Is Sorted	Medium	View Problem
10	3Sum	Medium	View Problem
11	Container With Most Water	Medium	View Problem
12	Trapping Rain Water	Hard	View Problem

Pattern 3: Sliding Window

#	Problem Name	Difficulty	LeetCode Link
13	Best Time to Buy and Sell Stock	Easy	View Problem
14	Longest Substring Without Repeating Characters	Medium	View Problem
15	Longest Repeating Character Replacement	Medium	View Problem
16	Permutation in String	Medium	View Problem
17	Minimum Window Substring	Hard	View Problem
18	Sliding Window Maximum	Hard	View Problem

Pattern 4: Stack

#	Problem Name	Difficulty	LeetCode Link
19	Valid Parentheses	Easy	View Problem
20	Min Stack	Medium	View Problem
21	Evaluate Reverse Polish Notation	Medium	View Problem
22	Generate Parentheses	Medium	View Problem
23	Daily Temperatures	Medium	View Problem
24	Car Fleet	Medium	View Problem
25	Largest Rectangle in Histogram	Hard	View Problem

Pattern 5: Binary Search

#	Problem Name	Difficulty	LeetCode Link
26	Binary Search	Easy	View Problem
27	Search a 2D Matrix	Medium	View Problem
28	Koko Eating Bananas	Medium	View Problem
29	Find Minimum in Rotated Sorted Array	Medium	View Problem
30	Search in Rotated Sorted Array	Medium	View Problem
31	Time Based Key-Value Store	Medium	View Problem
32	Median of Two Sorted Arrays	Hard	View Problem

Pattern 6: Linked List

#	Problem Name	Difficulty	LeetCode Link
33	Reverse Linked List	Easy	View Problem
34	Merge Two Sorted Lists	Easy	View Problem
35	Linked List Cycle	Easy	View Problem
36	Reorder List	Medium	View Problem
37	Remove Nth Node From End of List	Medium	View Problem
38	Copy List with Random Pointer	Medium	View Problem
39	Add Two Numbers	Medium	View Problem
40	LRU Cache	Medium	View Problem
41	Merge K Sorted Lists	Hard	View Problem
42	Reverse Nodes in K-Group	Hard	View Problem

Pattern 7: Trees

#	Problem Name	Difficulty	LeetCode Link
43	Invert Binary Tree	Easy	View Problem
44	Maximum Depth of Binary Tree	Easy	View Problem
45	Diameter of Binary Tree	Easy	View Problem
46	Balanced Binary Tree	Easy	View Problem
47	Same Tree	Easy	View Problem
48	Subtree of Another Tree	Easy	View Problem
49	Lowest Common Ancestor of a BST	Medium	View Problem
50	Binary Tree Level Order Traversal	Medium	View Problem
51	Binary Tree Right Side View	Medium	View Problem
52	Count Good Nodes in Binary Tree	Medium	View Problem
53	Validate Binary Search Tree	Medium	View Problem
54	Kth Smallest Element in a BST	Medium	View Problem
55	Construct Binary Tree from Preorder and Inorder	Medium	View Problem
56	Binary Tree Maximum Path Sum	Hard	View Problem
57	Serialize and Deserialize Binary Tree	Hard	View Problem

Pattern 8: Heap / Priority Queue

#	Problem Name	Difficulty	LeetCode Link
58	Kth Largest Element in a Stream	Easy	View Problem
59	Last Stone Weight	Easy	View Problem
60	K Closest Points to Origin	Medium	View Problem
61	Kth Largest Element in an Array	Medium	View Problem
62	Task Scheduler	Medium	View Problem
63	Design Twitter	Medium	View Problem
64	Find Median from Data Stream	Hard	View Problem

Pattern 9: Backtracking

#	Problem Name	Difficulty	LeetCode Link
65	Subsets	Medium	View Problem
66	Combination Sum	Medium	View Problem
67	Permutations	Medium	View Problem

68	Subsets II	Medium	View Problem
69	Combination Sum II	Medium	View Problem
70	Word Search	Medium	View Problem
71	Palindrome Partitioning	Medium	View Problem
72	Letter Combinations of a Phone Number	Medium	View Problem
73	N-Queens	Hard	View Problem

Pattern 10: Tries

#	Problem Name	Difficulty	LeetCode Link
74	Implement Trie (Prefix Tree)	Medium	View Problem
75	Design Add and Search Words Data Structure	Medium	View Problem
76	Word Search II	Hard	View Problem

Pattern 11: Graphs

#	Problem Name	Difficulty	LeetCode Link
77	Number of Islands	Medium	View Problem
78	Clone Graph	Medium	View Problem
79	Max Area of Island	Medium	View Problem
80	Pacific Atlantic Water Flow	Medium	View Problem
81	Surrounded Regions	Medium	View Problem
82	Rotting Oranges	Medium	View Problem
83	Course Schedule	Medium	View Problem
84	Course Schedule II	Medium	View Problem
85	Word Ladder	Hard	View Problem

Pattern 12: Dynamic Programming - 1D

#	Problem Name	Difficulty	LeetCode Link
86	Climbing Stairs	Easy	View Problem
87	Min Cost Climbing Stairs	Easy	View Problem
88	House Robber	Medium	View Problem
89	House Robber II	Medium	View Problem
90	Longest Palindromic Substring	Medium	View Problem
91	Palindromic Substrings	Medium	View Problem

92	Decode Ways	Medium	View Problem
93	Coin Change	Medium	View Problem
94	Maximum Product Subarray	Medium	View Problem
95	Word Break	Medium	View Problem
96	Longest Increasing Subsequence	Medium	View Problem

Pattern 13: Dynamic Programming - 2D

#	Problem Name	Difficulty	LeetCode Link
97	Unique Paths	Medium	View Problem
98	Longest Common Subsequence	Medium	View Problem
99	Edit Distance	Medium	View Problem
100	Burst Balloons	Hard	View Problem

Summary by Difficulty

Difficulty	Count
Easy	20
Medium	66
Hard	14
Total	**100**

Study Tips:

1. Master one pattern at a time before moving to the next.
2. Start with Easy problems to build confidence, then progress to Medium and Hard.
3. Understand the underlying concept rather than memorizing solutions.
4. Practice explaining your approach out loud as if in an interview.
5. Revisit problems after a few days to reinforce learning.